PRESIDENT'S OFFICE REGIONAL ADMINISTRATION
LOCAL GOVERNMENT.

BIOLOGY 01 F.4

MARKING GUIDE MARCH 2026

SECTION A.

1.

COLUMN A	(I)	(II)	(III)	IV	V	VI	VII	VIII	IX	X
COLUMN B	A	A	A	C	E	B	E	A	C	D

@ 1 = 10 marks

2.

COLUMN A	(I)	(II)	(III)	(IV)	(V)	(VI)
COLUMN B	G	I	E	K	H	C

@ 1 = 06 Marks

SECTION B

3. (a) Expected points from Yohana (Differences)

Animal cell	Plant cell
(i) Has no cell wall	Has cell wall
(ii) Has small temporary Vacuole	Has large permanent vacuole.
(iii) Has no chloroplast	Has chloroplast.
(iv) It is irregular in shape	Has regular shape.
(v) Has centrioles	Has no centrioles.
(vi) They are usually smaller	They are normally larger.

A student should ~~should~~ provide 3 points @ 1

(a) Expected points From George (Similarities between Plant and Animal cell).

(i) Both have Cytoplasm
(ii) Both have cell membrane.
(iii) Both have Nucleus.
(iv) Both have mitochondria.

Any 3 points @ 1 mark

3 (b) Importance of cell differentiation.
(i) It helps in Division of labour of the cell and organs.
(ii) It makes the body to work effectively through Coordinated organs.
(iii) It brings efficiency of the work.
Any two points @ 1.5 marks

4 (a) A child may suffer from Cholera. (2 marks)

(b) Caused by Bacteria called Vibrio cholerae and transmitted through Ingesting Contaminated food and water or through Vector Such as Housefly. (02 marks)

(c) Prevention of Cholera.
(i) Boiling or treating of drinking water.
(ii) Covering food.
(iii) Washing hands before and eating and after visiting toilets.
(iv) Washing fruits and vegetables before eating
(v) Prepare food in hygienic manner.
(vi) Proper use of clean toilet
(vii) Seek Immediately for medical treatment.
Any five points @ 1 mark = 5 marks.

5 (a) (i) A. Efferent arteriole.
B. Afferent arteriole.
C. Glomerulus.
D. Bowman's capsule.
(@ 1 = 04 marks)

5(a)(ii) Structure A is wide to maintain high blood flow into the glomerulus for filtration, and Structure B is narrow to restrict outflow of blood, increasing pressure in the glomerulus for effective filtration. (2marks)

(b) (i) It helps to remove waste product and toxic materials eg Urea, Carbon dioxide gas etc.
(ii) It eliminate the excess materials from our bodies like Soluble vitamins and drugs.
(iii) It Mantains the pH of the body fluids by removing excess bile pigment through liver.
(iv) It Mantains water balance in the body. Excess water is removed through sweating or urination.
(v) Regulation of Blood pressure by removing excess Salt and water in the body.
(vi) It gives chance for absorption of other materials.
Any 3 points. @ 1 = 03 marks

6(a) (i) oestrogen Hormone.
(ii) Progesterone Hormone.
(iii) Prolactin Hormone.
(iv) oxytocin Hormone.
(v) Luteinizing Hormone.
@ 1 mark = 5marks

6. (b) Similarities Between Meiosis and Mitosis.

(i) Both Involve diploid parent cells.

(ii) Both Involve Interphase, Prophase, Metaphase, anaphase and Telophase.

(iii) In Metaphase Individual chromosomes line up along the equator.

(iv) Both are types of cell division In which one cell give rise to more than one cell.

(v) Both require Spindle fibres to pull chromosome In opposite poles.

Any Four points (@ 1 mark = 4 marks)

7(a) How HIV destroy T-helper cell

(i) It reproduce Inside the T-helper cell then - rupture the cell membrane and new viruses are released.

(ii) It alter the T-helper cells so that when - it respond to an Infection it kills itself, Instead of dividing to form ~~an~~ a new cell.

(iii) It marks the T-helper cells as the target for destruction by other cells In the Immune System.

(iv) It causes fusion of many T-helper cells to form a giant cell. A giant cell can survive but it cannot perform the normal T-helper - cell function.

7 (b) Importance of blood circulation.

(i) It helps in transportation of nutrients eg glucose
(ii) It helps in transportation of oxygen gas.
(iii) It ensures the removal of waste products.
(iv) It helps in Regulation of body temperature.
(v) It helps in transportation of hormones from glands to the targeted area.
(vi) It helps in transportation of Antibodies

8 (a) Mendel's first law state that "Characteristics of an organism are controlled by internal factor (gene) which in pairs and only one of the factors will be carried in a single gamete" (01 mark)

(b) Let "R" represents gene for Red seed coat
"r" represents gene for White Seed coat.

Parental Phenotypes Red seed coat × White Seed coat
Parental Genotypes RR rr
meiosis
Gametes R R r r
Fertilisation
(02 marks)
F1 generation Genotypes Rr Rr Rr Rr
F1 generation phenotypes Red Red Red Red

8(b) Selfing of F_1 generation to obtain F_2 generation.

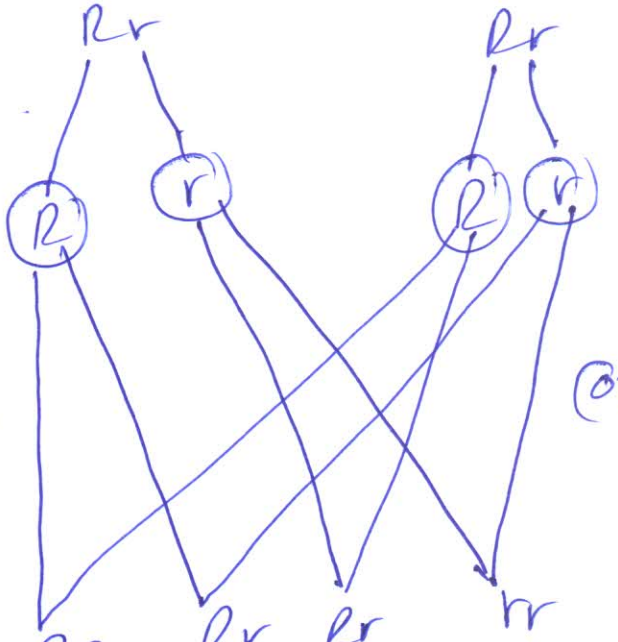
Parental Phenotypes Red Seed coat × Red Seed coat

Parental Genotypes Rr Rr

Meiosis

Gametes R r R r

Fertilisation



(02 marks)

F_2 generation Genotypes RR Rr Rr rr

F_2 generation Phenotypes Red Red Red White

(i) Genotypes of F_2 generation:

1 is RR, 2 are Rr and 1 is rr (01 mark)

(ii) Red Seed

$\frac{3}{4} \times 7324 = 5493$

Total number of Red Seed are 5493 (02 marks)

(iii) Percentage of Red Seed

$\frac{3}{4} \times 100\% = 75\%$ (01 mark)

(iv) percentage of white seed

$\frac{1}{4} \times 100\% = 25\%$ (01 mark)

SECTION C

09 Importance of Human Skeleton

(i) Support - It provide rigid frame work which support soft part of the body.
(ii) Movement
(iii) Protection of delicate Organs.
(iv) Formation of Red blood Cell
(v) It gives a definite shape
(vi) It store minerals such as Calcium and phosphorus

Introduction - 1½ marks

Main body - any 5 points @ 2½
Total - 12.5 or 12½

Conclusion - 1 mark.

10 Application of anaerobic respiration in our daily life

(a) Beer brewing (Brewery in industries) manufacture of bear and wine
(b) Bread making; leaving of bread using Yeast.
(c) Production of bio-gas for cooking and lighting
(d) Making of compost manure
(e) Production of vinegar (acetic acid)
(f) Production of power alcohol used to drive engenes.
(g) Production of silage (Animal food)

Introduction - 2 Marks
Any 6 points @ 2 marks
Conclusion 1 Mark

11 IN COLD ENVIRONMENT

(i) Increasing metabolic rate
(ii) Shivering
(iii) Vasoconstriction
(iv) Contraction of hair erector muscles

IN WARM ENVIRONMENT

(i) Vasodilation
(ii) Sweating
(iii) Relaxation of hair erector muscles
(iv) Decrease of Metabolic rate
(v) Panting and licking
(vi) Body extremities

Introduction – 01½ Marks
Main body – Any six ponts @ 02 Marks
Conclusion – 01½ Marks